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Processing Components in Song Attunement: A Modular Approach

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Abstract

Musical textsetting is the cognitive process by which we associate a text to a tune to perform a song, or by which we judge if a text and a tune match or not. Music cognition typically integrates the treatment of meter, rhythm, and pitch contour. The part of speech that is relevant for textsetting is prosody, including linguistic tones or pitch accents. This paper proposes a modular theory of musical textsetting: music and text are conceived as computed separately before being associated by means of *correspondence grammars*, which associate musical groups with prosodic phrases, stress with musical meter, and music tones with linguistic tones, and so forth.

Key words: musical textsetting, song processing, modularity, correspondence grammars, text-to-tune association

In 1993, musicologists Halle and Lerdahl published “A Generative Textsetting Model,” a paper developing an algorithm to predict the way a text can be associated with music in English folk songs. This work was continued by Halle (1999, 2005, 2008).

In 2003, neuropsychologists Peretz and Coltheart proposed a rich modular model of music, including connections between structural features and emotions, memory and the sensory-motor system. But their model was poorly developed concerning song. It contained a module called “vocal plan formation,” associated with the rest of musical structure by only one correspondence grammar.

Linguists, on their side, worked in three different directions.

Most of them followed the path opened by Halle and Lerdahl (1993) and tried to identify rules or constraints involved in “musical textsetting” (i.e., the association of texts to tunes). These linguists generally worked on specific languages and traditions (Aroui, *in press*; Chouvion et al., 2021; deCastro-Arrazola, 2015, 2018, p. 117-153, deCastro-Arrazola et al., 2015; Dell, 2011, 2015; Dell & Halle, 2009; Hayes, 2008, 2009a, 2009b; Hayes & Kaun, 1996; Hayes & MacEachern, 1996, 1998; Hayes & Schuh, 2019; Katz, 2023; Kiparsky, 2006; Premat et al., 2021; Proto, 2018; Proto & Dell, 2013; Rodríguez-Vázquez, 2006, 2006–2007, 2010a, 2010b, 2010c, 2011, 2013; Rossi, 2012, 2015; Ryan, 2022; San & Turpin, 2021; Temperley & Temperley, 2013; Turpin 2007a, 2007b; Turpin & Laughren, 2013). See also pioneering works, published before the seminal study by Halle and Lerdahl (1993), including Dell (1989 [1975]), Chen (1983), Morgan and Janda (1989, p. 278-285) and Palmer and Kelly (1992, p. 527-529). Proto (2015) provides an introduction to the field. These authors proposed different models, which often mix in a same set purely musical rules or constraints with textsetting rules or constraints. For example, Hayes and MacEachern (1998, p. 490-492) or Rodríguez-Vázquez (2010b, p. 199-200, 261) rank `FILLSTRONG`, a purely musical constraint (required to link preferably a strong beat to a musical event), within a same set than `MATCHSTRESS`, a textsetting constraint (required to link stressed syllables to strong beats).¹ The same problem occurs in Hayes (2009a), McPherson and Ryan (2018, p. 141sq.), and San and Turpin (2021, p. 6-8), among other works. This is not consistent with a modular approach of cognition, where cognitive components must be considered only in input/output interface relationships, and kept “encapsulated” (Fodor 1983, p. 69-83, Scheer 2011, p. 398-399, 435-444). In this paper, I distinguish music, language prosody, and musical textsetting as three distinct cognitive objects processed differently.

Other linguists have developed a quite independent research area concerning the correspondence of linguistic tones (or pitch accents) with musical tones. About 70% of the world’s languages are tone languages (Yip, 2007, p. 229). In these languages, a change of tone in a word or a phrase changes its meaning. How do singers attune the linguistic tones with the melody while preserving the meaning? Among the researchers dedicated to this problem, see Chan (1987), Saurman (1999), Wong and Diehl (2002), Wee (2007), Schellenberg (2009, 2012), Ketkaew and Pittayawat (2014), Proto (2016), Cho (2017), McPherson and Ryan (2018), Schellenberg and Gick (2020), Franich and Lendja Ngnemzué (2021), Kirby (2023). Ladd & Kirby (2020) provide an introduction to this field, which includes an extensive bibliography.

A few linguists challenge this way of thinking and emphasize that language processing and music processing may overlap in some ways (Dilley & McAuley, 2012,

1 Bruce Hayes (personal correspondence) tells me he would agree with this.

p. 27-29; Katz, 2022; Katz & Pesetsky, 2011, Schreuder, 2006, p. 5-31). In disciplines apart from linguistics, various researchers argue for this position. Some neuropsychologists (Bigand & Tillmann, 2020, p. 85-90; Peretz & Kolinsky, 2009, p. 143-144, 150, 159) propose that some modules may be dedicated to activities like rhythm processing, and these activities may be found in language as much as in music. Perrachione et al. (2013) led experiments suggesting shared cognitive processing of pitch in language and music. Patel (2008, p. 233-238, 2012a, p. 216-218) developed “the melodic contour deafness hypothesis,” according to which some brain zones are dedicated to verbal intonation as well as musical melodic contour. Patel also proposed a “shared syntactic integration resource hypothesis” (2008, p. 282-297, 2012a, p. 207-216, 2012b; Fedorenko et al., 2009): while different in their representations and syntactic domains, language and music would share the same neural resources to activate syntactic representations. It is also known that music perception may be influenced by the speaker’s native language (Ball, 2010, p. 360; deCastro-Arazola, 2018, p. 157; London, 2012, p. 58; Patel, 2008, p. 159-176; Peretz, 2012, p. 259, 262; Schreuder, 2006, p. 40). The most obvious feature that music and language share is recursion (cf. Hauser & Watumull, 2017; Lerdahl & Jackendoff, 1996 [1983], p. 16, Patel, 2008, p. 267, Roberts, 2012, p. 60-63). This questions the independence usually accepted between language and music in textsetting studies.

A related field of research is the question of the speech-to-song illusion (and its cousin the sound-to-music illusion), by which the simple repetition of a phrase produces for the hearer an effect of song (Deutsch, 2019, p. 151-169; Deutsch et al., 2011; Simchy-Gross & Margulis, 2018). Up to now, the experimental results do not provide conclusive evidence for shared or separated language and music processing (see Kubit et al., 2024, and Marjeh et al., 2024, for short overviews and full references).

One of the problems is to understand how linguistic sounds and music sounds, among various natural sounds, are perceived (and eventually produced). Two EEG-based experiments (Zuk et al., 2020) revealed that responses produced by the brain to speech and music sounds are stronger than the responses to other natural sounds; speech and music sounds get a better classification than other natural sounds in perception. But how is music distinguished from speech? Recently, two important fMRI experiments revealed that brain regions dedicated to music processing do not overlap with those devoted to language processing (Chen et al., 2023), and that there exists “a neural population selective for song in human auditory cortex” (Norman-Haignère et al., 2022). Significantly, this neural component has a relatively long latency (Norman-Haignère *et al.*, 2022, p. 1474). This may suggest a *textsetting system*, activated *after* language processing and music processing.

Most of the studies devoted to the textsetting develop, explicitly or implicitly, a modular approach to textsetting and envision music and language as forming two completely independent modules. However, their underlying theoretical assumptions are not always very clear. The question is to know how modules interact and participate in producing or perceiving well-formed text-to-tune associations. There is an urgent need for a coherent theoretical approach.

The present paper aims to make explicit the processing components of textsetting. Departing from Peretz and Coltheart (2003), it tries to formulate a consistent theoretical framework for research devoted to the textsetting problem. The aim is to identify the grammars involved in musical textsetting, leaving aside emotions and the interface with the sensory-motor system. The term *grammar* is used in a way that is widespread among linguists; it designates a cognitive processing system. The objective is to have a coherent modular approach, and to separate constraints belonging to different grammars (modules or correspondence grammars). The proposed model is consistent with Chen et al.'s (2023) and Norman-Haignère et al.'s (2022) findings. It predicts structural features in different types of monophonic music: clapping monophony (§.2), melodic instrumental monophony (§.3), chant (§.4), and monodic song (§.5). More complex situations, in homophonic or polyphonic music, are not envisioned (in *polyphonic* music, two or more melodies are superimposed on one another, while in *homophonic* music a same melody includes chords). The global model is presented and discussed in §.6 and §.7, while the idea of a competition between grammars is briefly introduced in §.8, before the conclusion in §.9.² This model does not rule out research that looks for overlaps between music processing and language processing, but it formulates explicit hypotheses which could inform the design of new experiments and guide further research.

1. Musical Meter: Grid-Building Constraints

To begin, we must envision the form of the module concerning the conditions of well-formedness of abstract meters.

In a music grammar, there is generally a set of constraints dedicated to the mental processing of meter. Formally, a meter is represented by a metrical grid (these are musical grids, and not the verbal grids used by phonologists, or by Schuh, 2011). It may seem strange to envision constraints on the metrical grids alone, as a meter does not exist independently of rhythmic events. But, in the same way as syntactic trees, which have their own properties in linguistic theory, abstract metrical grids have their own constraints. A musical meter is made up of *beats*, also called *metrical positions* (and represented by crosses) and beats are separated by *time-spans* (represented by blank

2 Kiparsky (2006, p. 7) proposed a modular model of song containing three tiers: musical rhythm, linguistic prominence and poetic meter. The hypothesis of a poetic meter in song can be challenged, but deserves an independent study. I cannot discuss it in this paper. Any debate on this topic should integrate Saito et al.'s (2006) findings. They compared neural activity for a same Japanese text depending on whether it is sung or spoken. The text was composed of phrases containing a regular number of moras (about moras, see §.4.1), five or seven, as in Japanese poetry. It is significant to notice that neural regions activated for spoken versions were just a subpart of those activated for sung versions. This result may suggest that the text without music includes nothing more than the same text performed with music. In other words, one may think that it does not integrate any prosodic meter. Nevertheless, Saito et al.'s research cannot help to reject decisively three other hypotheses: 1) the two versions contain a prosodic meter (and the sung version has two meters: the musical one and the linguistic one); 2) both versions contain one single metrical structure (which is musical but becomes impoverished in the spoken version); 3) a prosodic meter is only a linguistic object.

spaces between crosses). Some beats are stronger than others. In a grid, each level of strength is represented by a specific horizontal row of crosses (a *gridline*).

A metrical grid predicts that two different phenomena are mixed in meter perception: equal time-spans are perceived between beats (this is represented in the grid by gridlines), and a specific *strength* or *salience* is assigned to each beat (it is represented by columns). Strictly speaking, the strongest beats are not necessarily the most salient ones, as our perception system seems to favor intermediate levels of metrical structure (see Lerdahl & Jackendoff 1996 [1983], p. 21). In this article, we will ignore metrical levels that are above the most salient ones.

In example (1), numbered columns of crosses represent the beats themselves. The more crosses there are in these columns, the stronger the beat. Beat 1 is stronger than beat 3, which is stronger than beat 2. Crosses belonging to a same gridline are separated from each other by equal lengths, representing equal time-spans. When a beat is stronger than the previous one, it is said to be *projected* to the upper level. A beat from gridline *a* which is not projected to the upper level is a *weak* beat. A beat projected from gridline *a* to gridline *b*, and only gridline *b*, is *medium*. If it is projected also to gridline *c*, it is *strong*. Medium beats and strong beats are *marked beats*. A grid having two rows has weak beats and strong beats only. In a 4-rows grid, we will assume that gridline *d* represents *extrastrong beats*. There may also be local *extraweak beats*, indicated on a gridline labeled 0. These are necessary, for example, to represent the local metrical structure of a triplet. Theoretically, rows and columns in a grid are potentially infinite. Practically, the horizontal grid length and the number of represented columns are those relevant for perception.

(1) *The caba meter*

	1	2	3	4	5	6	7	8
c	x				x			
b	x		x		x		x	
a	x	x	x	x	x	x	x	x

In example (1), the meter is made of four beats. Two of them are weak, one is medium, and one is strong. The sequence *strong-weak-medium-weak* appears for the first time positions 1 to 4 and then is repeated; it is the meter. I propose to call this meter *caba* as its first position is projected to gridline *c*, its third position is only projected to gridline *b* and the other positions are weak (they are only represented gridline *a*). A meter *must* be (mentally) repeated to be perceived as such.

Lerdahl and Jackendoff (1996 [1983]) formulated some of the principles and constraints holding for a metrical musical grid. For example, the technical principle (2) is necessarily respected in all metrical music traditions.

- (2) MWFR2. Every beat at a given level must also be a beat at all smaller levels present at that point in the piece (Lerdahl & Jackendoff 1996, [1983], p. 72).

According to this principle, grid *(3) is ill-formed, because there is no beat represented in a7 and b5 (starred examples are ill-formed or unpreferred).

* (3)	1	2	3	4	5	6	7	8
c	x				x			
b	x		x				x	
a	x	x	x	x	x	x		x

In all metrical music traditions, two beats represented at a same level n can only be separated by one or two beats from level $n-1$. This inviolable constraint is expressed in (4).

- (4) Exactly one or two beats at a given level must elapse between each pair of beats at the next level up. (Temperley, 2004, p. 37).³

According to (4), grids (5) and (6) are well-formed, but grids *(7) and *(8) are ill-formed.

(5) *The cababa meter*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
c	x						x						x					
b	x		x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

(6) *The caabaa meter*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
c	x						x						x					
b	x			x			x			x			x			x		
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

* (7)		1	2	3	4	5	6	7	8	9	10	11	12
c		x				x				x			x
b		x	x			x	x			x	x		x
a		x	x	x	x	x	x	x	x	x	x	x	x

* (8)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c		x								x							
b		x				x				x				x			
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

(4) is universal and always inviolable in two aspects: a projection from gridline n to gridline $n+1$ is always minimally binary and always maximally ternary. It is never unary, as it is between positions 1–2, 4–5, 7–8 and 10–11 in the transition from gridline a to gridline b in *(7). Unary projections never exist in musical grids. *(8) is ill-formed because time-spans in gridline b are four times longer than those in gridline a : they are quaternary projections. For perceptual efficiency, the mind prefers to use binary positions and to add a metrical level, as in (9).

3 Lerdahl and Jackendoff's (1996 [1983], p. 69) MWFR3 is slightly different.

(9) *The dabacaba meter*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
d	x								x							
c	x				x				x				x			
b	x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

The strength of a constraint may change from one musical idiom to another. For example, constraint (10) seems almost never violated in Western classical tonal music.

- (10) MWFR4. Each metrical level must consist of equally spaced beats.
(Lerdahl & Jackendoff 1996 [1983], p. 69).

Grids are not relevant to represent ametrical music and need some special amendments to represent phenomena such as the *accelerando*. Formal studies of ametrical sung traditions are found in Chen (1983) and Drescher (2008); see also Clayton (1996). In Western classical music, in a given gridline, beats are generally equally separated in time.

Some musical idioms, for example the Balkan folk tradition (Arom, 2004; Ball, 2010, p. 212; Hannon & Trehub, 2005), know *additive signatures* (London, 2012, p. 89), which commonly do not respect (10). In Greek folklore a common meter is 2+2+3/4 (Jackendoff, 1987, p. 222). In jazz, the meter of Paul Desmond's *Take five* is 3+2/4. Generally, these tunes seem to alternate two different meters. Constraint (4) is not violated, as 5/4 is more appropriately described as an alternation of 3/4 and 2/4 (as in *Take five*), and 7/4 as an alternation of 4/4 with 3/4; 4/4 is made with strictly binary projections, as in (1) or (9).

2. Musical Rhythm and Meter-to-Rhythm Synchronization

2.1. Rhythm and Rhythmic Groups

Rhythmic events⁴ are not necessarily linguistic units, and they do not necessarily have a specific pitch, although in many drum traditions around the world, the pitch of the stroke may be partly relevant. Handclaps might be a perfect example of rhythmic events for a musical piece. The only part of a rhythmic event that is appropriate in clapping music is its attack point. Below, the attack point of a rhythmic event is represented by the sign “●.” Time spans between attack points are called interonset intervals (IOI).

Rhythmic events tend to form groups. The mental perception of groups is based on principles known since the Gestalt psychology of the 1920s. The two most important principles are similarity and proximity (Ball, 2010, p. 139-144; Deutsch, 2013, p. 183-184, 196; Kanizsa, 1997 [1980], p. 19-24; Katz, 2022, p. 4-6, 21-23). In terms of musical rhythm, two events can belong to the same group if they are perceived as sufficiently similar. Two handclaps, for example, sound similar enough to potentially belong to a same group. Concerning proximity, the more two consecutive rhythmic events are distant from each other in time, the less chance they have of belonging to a

4 Equivalent to rhythmic events are gesture events, for example in dance or in Sign Languages. I will not develop this point.

same group. (11) is a procedure adapted from Aroui (in press) to identify rhythmic groups.

(11) *Procedure to discover rhythmic groups*

Step 1 (Gaps): Place a group boundary between two attack points: A) when an IOI is larger than the previous IOI; and/or B) when an IOI is larger than the following IOI.

Step 2 (No Unary): Avoid rhythmic groups formed by a single rhythmic event. A single rhythmic event is necessarily associated with the group that precedes it or the group that follows it.

Step 3. (Parallelism): A) Prioritize the groups that are repeated. B) A sequence of groups that is repeated forms a group itself, provided that it does not exhaust the rhythmic events of the piece.

(adapted from Aroui (in press))

In (11) we have an original three-step procedure, each of the steps allowing the elimination of a certain number of candidates from the rhythmic group of the previous step. *Gaps* and *No Unary* distinguish group boundaries that are temporally relative: an IOI forms a group boundary when it is larger than IOIs that precede and/or follow it. As it is formulated, this procedure does not specify the exact interval of time associated with the IOI under consideration. The longer an IOI is, the more likely it is to match a rhythmic group's boundary.

According to (11), there are two musical groups in (12), where digits express a regular temporal scale. One is associated with time positions 1–3, the other with the string of positions 5–7. Distinguishing these musical groups is important for textsetting studies. But it is clear that these groups exist independently of an eventual text set to the music.

(12)	1	2	3	4	5	6	7	8
R	•	•	•		•	•	•	

On row R (*R* for *rhythm*), the sequencing of rhythmic events results in a temporal form that is called a *rhythm*.

2.2. Meter-to-Rhythm Correspondence Grammar

Strong clinical evidence for a cognitive dissociation between *meter* and *rhythm* has been brought by various observations and experiments: brain-damaged patients may be affected in the perception of rhythms, while their perception of meters does not differ significantly from the control group (Levitin; 2006, p. 169; Liegeois-Chauvel et al., 1998, p. 1865). A universally unviolated constraint concerning the setting of attack points to the grid is (13).

- (13) MWFR1. Every attack point must be associated with a beat at the smallest metrical level present at that point in the piece. (Lerdahl & Jackendoff 1996 [1983], p. 72).

Trills, appoggiatura, or other ornaments seem to be exceptions to (13). But they do not change the *structure* of the tune. Grid *(14) violates (13), because the third attack point is not associated with any beat in the grid.

*(14)	1	2	3	4	5	6	7	8
c		x				x		
b		x		x		x		x
a	x	x	x	x	x	x	x	x
R	•	•		•	•	•		•

In *(14) and subsequent grids, there are association lines below gridline *a*. Here they associate some metrical positions to the attack points of rhythmic events. They may also associate other modules, submodules or domains, as we will see.

(15) is a constraint concerning the choice of beats to be associated with attack points.

(15) MPR3. Prefer a metrical structure in which beats of level L_i that coincide with the inception of [sound-]events are strong beats of L_i . (Lerdahl & Jackendoff, 1996 [1983], p. 76).

The constraint FILL STRONG POSITIONS used and defined by Hayes and MacEachern (1998, p. 490) or Rodríguez-Vázquez (2010b, p. 200) is a variant of (15), adapted to their corpora. Because of (15), grid (16) is preferred over grid *(17).

(16)	1	2	3	4	5	6	7	8
b	x		x		x		x	
a	x	x	x	x	x	x	x	x
R	•	•	•		•	•	•	

*(17)	1	2	3	4	5	6	7	8
b		x		x		x		x
a	x	x	x	x	x	x	x	x
R	•	•	•		•	•	•	

According to another constraint, a sound event associated with a strong position tends to be “long”, (i.e. tends to be immediately followed by an empty metrical position, Hayes, 2009a).⁵ In music, a meter/rhythm association like (18a) seems more natural than (18b).

⁵ Temperley’s “length rule” (2004, p. 32) is something similar. Lerdahl and Jackendoff’s (1996 [1983], p. 84) rule MPR 5 is richer: it includes the interface of meter with other aspects of musical structure.

(18a)		1	2	3	4	5	(18b)		1	2	3	4	5
	c	x				x		c	x				x
	b	x		x		x		b	x		x		x
	a	x	x	x	x	x		a	x	x	x	x	x
	R	●		●	●	●		R	●	●	●		●

In (18a) and (18b), rhythmic events are associated with the same number of strong, medium, and weak positions. But in (18a) the rhythmic event associated with the first strong position is immediately followed by an empty position. This trend can be represented by the constraint **STRONG IS LONG**.

When two musical groups have a same form, they are similarly aligned with the meter. This constraint is called **POSITIONAL PARALLELISM** and is made explicit in (19).

- (19) **POSITIONAL PARALLELISM**. Subject to constraint (15), when two or more regularly repeated groups have a similar rhythm, they are aligned similarly on the meter. (Inspired by Lerdahl & Jackendoff's, 1996 [1983], p. 75, rule MPR1 and Temperley's, 2004, p. 70, PSPR 3 rule.)

In (16), a rhythmic group containing three sound events is repeated, and **POSITIONAL PARALLELISM** is respected. In *(20), on the other hand, the alignment does not respect this constraint: a rhythm is repeated, but it is not always associated with the meter in a same way. For example, the first sound event of the repeated rhythmic group is sometimes associated with a strong position (position 1) sometimes with a medium position (position 11), sometimes with a weak position (positions 6 and 16).

*(20)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
	c	x				x				x				x				x	
	b	x		x		x		x		x		x		x		x		x	
	a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
	R	●	●	●		●	●	●		●	●	●		●	●	●		●	●

Rhythmic variations remain possible on weak or extraweak positions (see Halle 2005, p. 51). To account for these complications, **POSITIONAL PARALLELISM** could be conceived as a family of constraints. One of the constraints in this family would apply on the weak metrical level, another one on the medium level, and so on (on constraint families, see Prince & Smolensky, 2004 [1993], p. 29-30 *et passim*). In certain musical traditions, constraint (19) is respected especially for groups of musical groups. Other variations are possible on the positions separating two groups (Halle, 2005, p. 26-27, 46).

(21) is another constraint, subordinated to (15). In certain cases, this constraint leads to the choice of one alignment over others.

- (21) **GROUP ONSETS**. Subject to what constraints (15) and (19) allow, the first higher-level position of a musical group appears as early as possible in

this group. (Inspired by Lerdahl & Jackendoff ‘s, 1996 [1983], p. 76, rule MPR2)⁶

In some traditions, this constraint is strong enough that a piece of music always begins on a strong position (there is no *anacrusis*). This is the case, for example, in the Warlpiri songs studied by Turpin and Laughren (2013, p. 409).

Depending of the way it is associated with the grid, a musical group may have a *masculine* or a *feminine* musical ending; see (22).⁷

- (22) *Feminine and masculine musical group endings (definition)*. A group ending is *masculine* if its last attack point is on a stronger position than the next to last; it is *feminine* if its last attack point is on a weaker position than the next to last.

(16) is an example of musical groups having masculine endings. (23) is an example of a feminine musical group.

(23)	1	2	3	4	5	6	7	8
b	x		x		x		x	
a	x	x	x	x	x	x	x	x
R	•		•	•	•	•		

Usually, the terms *masculine* and *feminine* are used for cadences. But cadences are composite objects that involve pitches (see below, §.3.4). The discrimination of *masculine* or *feminine* musical endings is a purely metric-rhythmic problem, which has nothing to do with pitches. The “gender” of musical endings is important when one studies the correspondence of rhythmic music groups with linguistic phrases (see §.4.4).

3. Tone Alignment in Monodic Instrumental Music

3.1. Tones and Tone Groups

In some cases, a musical sound event may have a regulated tone. This happens, for example, if the musical piece is performed with a flute. A music tone always has a specific pitch (a special location in a scale). More exactly, a music tone has a definite pitch and, like any natural harmonic sound, is made of *partial tones*, one of them being the *fundamental tone* and the others being *harmonics* (a.k.a. *overtones*).

⁶ See also rule MPR 4 (*Grouping Rule*) in Temperley (2004, p. 38).

⁷ The terminology that opposes *feminine* to *masculine* endings is very old and of linguistic origin. It was already used by Jacques Legrand before 1405 (Kastner, 1904, p. 11). It has its source in the morphology of Middle French, a language where morphologically feminine words added a post-tonic *-e* [ə] at the end of the word; this *-e* is still found in the spelling of Modern French (*étudiant/étudiant+e*, *joli/joli+e*, etc). The post-tonic [ə] was called *feminine* because it was found mainly in the feminine forms of nouns and adjectives. The feminine ending is said to be weak and the masculine ending to be strong because the [ə] is post-tonic and the masculine forms are oxytones (stressed on their last syllable). Rhythmically, these endings are parallel to *masculine* and *feminine* group endings in music.

Being optional and processed independently, music tones have to be represented on a separate gridline. This independence of the tone component reveals that the notion of *note* (characterized by a rhythmic event and by a specific tone) is a composite notion, relevant for melodies and songs, but not for clapping music or for street chants. (25), a grid transcript of score (24), is an example of a grid in which the rhythmic events are associated with music tones. This folk dance is traditionally played on a little Provençal flute called a *galoubet*. The transcription is mine.

(24) *Farandoulo del Tarascaire (excerpt)*



(25)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
c		x						x						x						x					
b		x			x			x			x			x			x			x				x	
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P		Bb	C'	Bb	Bb	-	G	Bb	C'	Bb	Bb	-	G	G	G	G	G	F	Eb	F	Eb	D	Eb	-	-
R		●	●	●	●		●	●	●	●	●		●	●	●	●	●	●	●	●	●	●	●	●	●

In grid (25), row P (*P* for *pitch*) represents music tones (the row is not labelled T because T is going to be used for the Text). A tone label followed by a prime sign indicates that the tone is higher than the medium *B* in the stave. A tone is attached to a metrical position by its inception point.

The tonal contour of a musical sequence can produce the perception of musical groups. Two consecutive tonal events tend to be grouped together when they are close to each other in pitch (Ball, 2010, p. 216; Deutsch, 2013, p. 196; Patel-Grosz et al., 2023, p. 35). The resumption of a melodic contour produces an effect of tonal parallelism that leads to the perception of two musical groups (Halle, 2005, p. 42), even if the rhythmic structure does not lend itself to it. Conversely, a change in the melodic contour generates the perception of a switch in musical groups (Jackendoff & Lerdahl, 2006, p. 38-39).

The melodic contour can be preserved when the parallel tones are not identical: one of the two parallel groups can be overall higher than the other. This is the case at the beginning of the well-known French nursery rhyme *Sous le pont d'Avignon*, reported in (26) (the lyrics are ignored).

(26)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
b		x		x		x		x		x		x		x		x	
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P		G	G	G	-	A	A	A	-	B	C'	D'	G	F#	G	A	D
R		•	•	•		•	•	•		•	•	•	•	•	•	•	•

In (26), tonal events located at positions 9 to 12 are parallel to those associated with positions 13 to 16, although the tones are different. What matters is that in both cases

the melodic contour is the same: up to the third tonal event of the group, the curve rises, then it falls for the fourth.

Constraints which allow to form musical groups based on musical tones are proposed in (27).

(27) *Tone groups*

TONAL PARALLELISM. When a tonal contour is repeated, its edges are a musical group's edges.

TONAL CONTRAST. An unpeated tonal contour makes a musical group.

3.2. The Rhythm-to-m-tones Correspondence Grammar

There is a musical correspondence grammar that processes the interface between tones and rhythm. We can label this grammar *Rhythm-to-m-tones*. M-tones stands for musical tones (they will be opposed to *linguistic tones* in §.5). This grammar involves constraints (28) and (29).

(28) Every rhythmic event must be associated with one and only one tone event.

(29) Two rhythmic events may not be associated with the same tone event.

These constraints predict a one-to-one correspondence between rhythmic events and tonal events. It is crucial to distinguish the notion of *tone* and the notion of *tone event*. When two consecutive sound events have the same pitch, the rhythmic events are not associated with the same tone event: see, for example, positions 3 and 4 in (25). Constraint (28) may be violated when tones are not discrete, as in the case of *glissando*.

Rhythmic groups and tonal groups may be in conflict. Compare (30a) and (30b), which are adapted from Patel-Grosz et al. (2023, p. 36). In (30a) rhythm groups and tone groups coincide. In (30b), they differ; rhythm requires a group of three notes followed by a group of two notes, while melody necessitates the contrary (a group of two notes followed by a group of three notes).

(30) a.  b. 

Deutsch (2013, p. 210-211) reports a series of experiments which reveal that, in case of conflict, subjects tend to perceive groups based on rhythm (i.e., on temporal segmentation), as seems to be the case in (30b). But various factors may lead to the opposite result. Consider example (25), ignoring its metric structure. The melodic contour allows one to distinguish musical groups placed respectively on positions 1–6 and 7–12. Indeed, the tones located at positions 7 to 12 repeat in the same order the tones located at positions 1 to 6. When considering rhythm only, one would rather distinguish a musical group on positions 1 to 4, another on positions 6 to 10 and a third on positions 12 to 22, by virtue of procedure (11). However, in (25), the tones make possible the perception of different musical groups and they seem to prevail over the

rhythmic groups. In the same way, in (26), the tonal contour favors distinguishing two groups between positions 9 to 16 (see above §.3.1), while rhythmically there is just one group.

There is a lot more to be said about musical groups. In addition to rhythm and tonal structure, it has been shown that factors such as timbre, tempo, intensity, sound continuity, and space (the places where sounds originate) play a role, as well as working memory. For an introduction, see Deutsch (2013). See also Temperley (2004, p. 332-333) and Patel (2008, p. 108, 190, 197).

3.3. Meter and Tones (Meter-to-m-tones Grammar)

Tonality is a regulated set of tones. It is universal (Mehr et al., 2019, p. 13-15) but its *modes* may vary from one tradition to another. Modern Western music distinguishes between a major mode and a minor mode. In the major mode, the octave is divided by the following intervals (digits indicate the number of semitones forming an interval): 2–2–1–2–2–2–1. The minor mode has the intervals 2–1–2–2–1–2–2. It seems that the unequal intervals help to situate notes within the chosen mode (Levitin, 2006, p. 34-35; Patel, 2008, p. 21). The first tone of a mode (the one before the first interval) is particular and is called the *tonal center* or *tonic* of the mode (Mehr et al., 2019, p. 13). Music with a tonic is tonal music. The tonic is the tone from which the other intervals are calculated.

Various modes exist across the world; some vary by the number of intervals, others by the size of these intervals (see Patel, 2008, p. 17-20). Pentatonic modes are particularly common. They distinguish, within the octave, five different tonal classes, instead of seven as in the modes of Western classical music. The blues, for example, are characterized by the intervals 3–2–2–3–2.

Modes may be used to make *scales*. A scale is a norm that decides which specific pitch class must be associated to the tonic. One might object that the modes of music of the Middle Ages and the Renaissance are also formed with a specific tonic. By tradition, it is said that the dorian has for tonic the *D* and that the notes are *D E F G A B C (D)*. Similarly, it is said that the tonic of Phrygian is *E* and that the notes are *E F G A B C D (E)*. But in fact, at the time, there was no universal tuning fork (Abromont & Montalembert, 2001, p. 341). The exact height of the tonic was therefore relative. What distinguishes modes, whether old or modern, are the intervals, and they share the need for a relative tonic. Scales are standards through which the tonic is no longer relative and is regulated by a tuning fork. In the scale of *C major*, the tonic is *C* and the intervals are the ones given in (31). The intervals characterizing the *A minor* scale are given in (32).

(31) Tones and intervals, C major scale

C	D	E	F	G	A	B	C'
	2 semitones	2 semitones	1 semitone	2 semitones	2 semitones	2 semitones	1 semitone

(32) Tones and intervals, A minor scale

A	B	C	D	E	F	G	A'
	2 semitones	1 semitone	2 semitones	2 semitones	1 semitone	2 semitones	2 semitones

The tones making up the *C major* scale and those making up the *A minor* scale are exactly the same. How is it possible for the listener to know if a piece of music made of these tones is in *C major* or in *A minor*? The solution relies notably on the frequency with which a given tone is associated with a strong metrical position (Levitin, 2006, p. 35-36). In homophonic music, chords are also important. Newborn infants are already sensitive to the difference between major chords and minor chords (Bigand & Tillmann, 2020, p. 50). In any case, the setting of tones on the musical meter helps to discriminate the major mode and the minor mode. I will label *Meter-to-m-tones* the grammar in charge of the association of tones with the metrical structure.

The peak and valley of a melodic contour may constitute melodic accents (Müllensiefen et al., 2009, p. 24). In (26), the high *D* and the low *D* (the tonal events numbered 11 and 16) carry a melodic accent. Melodic accents can reinforce the metric structure (when they are associated with strong positions) or oppose it (when they are associated with weak positions). This is another aspect of the *Meter-to-m-tones* grammar.⁸

3.4. The Grammar Associating Musical Tones with the Metric-Rhythmic Structure

Compare score/grid *(33/34)—a modified version of (24)—with score/grid (35/36). Consider *(33/34) as the end of a piece of music.

*(33) *Farandoulo del Tarascaire* (modified excerpt)



*(34)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
c	x						x						x					x						
b	x			x			x			x			x			x		x			x			x
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	Bb	C'	Bb	Bb	-	G	Bb	C'	Bb	Bb	-	G	G	G	G	G	F	Eb	F	Eb	D	E	-	-
R	•	•	•	•		•	•	•	•	•		•	•	•	•	•	•	•	•	•	•	•	•	•

(35) *Jue de la Pico* (excerpt)



8 If we consider as well the phenomenal accent (a greater intensity assigned to a musical event in the signal), we obtain a set of tripartite relationships between the metrical structure, the melodic accent, and the phenomenal accent, each of which being able to reinforce or oppose each of the other two. The produced effects are important and can, depending on the case, be perceived as stimulating (an interesting effect of surprise due to a discrepancy between two of these levels) or as clumsy (the discrepancy does not seem to be resolved, it leaves the listener with a feeling of dissatisfaction). These questions, which are close to the problem of musical emotions, are left aside in the present work.

(36)		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
c		x						x						x						x					
b		x		x		x		x		x		x		x		x		x		x		x		x	
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P		F'			E'	F'		Bb				C	D'	Eb'		D'		C'		Bb	A	G	F	Bb	C'
R		•			•	•		•				•	•	•		•		•		•	•	•	•	•	•

		25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
c		x						x						x						x					
b		x		x		x		x		x		x		x		x		x		x		x		x	
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P		F'			E'	F'		Bb				C	D'	Eb'		D'		C'		Bb					
R		•			•	•		•				•	•	•		•		•		•					

Why is *(33/34) ill-formed while (35/36) is not? In both cases some of the tones used are out of scale. *(33/34) is in *E flat* major, and its last *E* (position 22) is not flat. (35/36) is in *B flat* major, but the *E* in the first and the fifth measures are not flat (contrary to the *E*, measures 3 and 7).

To explain the difference in well-formedness between (35) and *(33), we have to introduce the notion of *cadence*. A *cadence* is a group of notes that mark the end of a musical motif (i.e., the end of a rhythmic-melodic group). It is a composite object that results from the interaction of tonal events with the whole metric-rhythmic structure. At a melodic level, the scale of a piece of music tends to be respected in cadences and, typically, a final musical group is masculine and ends with the *tonic* pitch of this scale, as in (35). Pitch alterations of the scale may happen commonly within a musical group when they do not belong to the cadence, as in the first and fifth measure of (35). Pitch alterations tend to occur more often in weak metrical positions, as in (35). *(33) sounds bad because its pitch alteration coincides with the resolution of the cadence and this cadence ends a piece of music. In Western classical music, *(33) would be possible at the end of a musical group that does not end the whole piece of music. This unexpected cadence would be corrected later by another cadence realizing the expected *E flat*.

More specifically, (35/36) is well-formed for the following reasons: 1) the *E* is on a weak position; 2) it is very short (it is immediately followed by a metrical position associated with a note); 3) the following note is an *F*, i.e., the dominant for the scale of *B flat* major; 4) this *F* is on a stronger position than the *E*; 5) properties 1, 2, 3, and 4, considered together, help to *anchor* the *E* onto the *F*: there is briefly a melodic *tension* because of the atonality brought by the *E*, but this tension is immediately resolved by the *F*. The *E* is a transitory melodic step leading to the *F*. (See Bigand & Tillmann, 2020, p. 132-133).

If *(33) features an illegitimate ending for a piece of music, it is also because its final *E* is the note harmonically furthest from the *B flat* which constitutes the tonic of this piece. There exists a *circle of fifths* in which the twelve constituent notes of the octave are placed on the circumference of a circle. Each note is followed by the one with which

it has a fifth interval. The further a note is from another on this circle, the more inharmonic is its relationship to it (see Patel, 2008, p. 251, 2012a, p. 210-211).

In the case of positional parallelism (see §.2.2), rhythm and melodic contour may be subject to variations on weak or extraweak positions. Melodic variations at the end of parallel groups are also frequent: the beginning of the melody is repeated (and there is a metric-rhythmic parallelism) but its end is subject to a variation (Jackendoff & Lerdahl, 2006, p. 39).

4. Textsetting Constraints in Chant

4.1. The “Text”

The text has its own grammar, whose rules and constraints are the concerns of linguists who study languages in their everyday use. The most important linguistic units involved in musical textsetting are prosodic constituents. They are defined and identified by the theory of prosodic hierarchy. This theory exists in different versions. I have found particularly useful for textsetting studies Selkirk (1984, 1986, 1996), Nespor and Vogel (2007 [1986]), Jackendoff (1987, p. 74-85), Selkirk and Tateishi (1988), Hayes (1989), Peperkamp (1997), Rialland (2003), Schreuder (2006), Labrune (2012a, p. 142-177, 2012b), Post (2012), and Kager and Martinez-Paricio (2019). One of the last versions of the theory is *Match Theory* (see Ishihara & Kalivoda, 2022). An alternative is Direct Interface Theory (see Scheer, 2011). The theory of prosodic hierarchy is generally understood as supporting modularity theory (Selkirk, 1996, p. 189). It assumes a limited set of prosodic constituents; these constituents covering the gamut of possible prosodic forms, from the smallest to the largest. The closed list of these constituents varies somewhat from one variant of the theory to another. The list of prosodic constituents in (37) may be appropriate for the study of musical textsetting.

(37) *Hierarchy of Prosodic Constituents*

mora (μ) < syllable (σ) < phonological foot (Σ) < phonological word (ω)
< phonological phrase (φ) < intonational phrase (IP) < utterance (U)

A syllable may end in a vowel or a consonant. The vowel is its *nucleus* and the consonant its *coda*. A *mora* is a prosodic unit that is relevant in languages sensitive to weight. In many of these languages, a syllable containing a long vowel or ending in a consonant has two moras: from a moraic point of view, a long vowel equals a short vowel followed by a consonant. If the vowel is short and the coda empty, the syllable contains only one mora. The remaining languages sensitive to weight divide into distinct categories: those where only the vowel length is relevant for moras; and those where this is just the presence or absence of a consonant in the coda that is relevant. Lastly, in a language like Japanese, the mora is central and the role of the syllable is debatable (Labrune, 2012b).

The phonological foot is the core unit for rhythm in prosody, as it is typically made up of a sequence of syllables (or moras), one of them being stronger than the others (for example because it bears a stress). The phonological foot will not be used directly in this work, as I shall focus on stresses or accents.

A phonological word expresses itself differently from one language to another. Roughly, it is made up of one or more phonological feet. Usually, one of the feet situated at the edges of the word is stronger than the others (Nespor & Vogel, 2007, [1986], p. 142). The phonological word may include affixes, clitics, and various particles. It is often assumed that at least one of its edges must coincide with the edge of a syntactic constituent (Selkirk, 1986, p. 383-397, 1996, p. 190-192). In English, “A chair” is a phonological word, as well as “stainless”. Compounds are a difficult matter for the phonology of words. They behave differently from one language to another and even sometimes within a same language (Peperkamp, 1997, p. 107-108, 117-155).

In English, the phonological phrase is the domain within which *Iambic Reversal* applies (Nespor & Vogel, 2007 [1986], p. 55-56, 177-178). This operation is also called *stress shift* or *rhythm rule* (Liberman & Prince, 1977, p. 255). The phonological word *sixteen* is normally stressed on its last syllable. But in the phonological phrase “sixteen tones,” this stress shifts to the first syllable, to avoid a stress class with the word “tones.”

The intonational phrase (IP) is quite important for musical textsetting. It has a falling or a rising contour and may be followed by a pause (Nespor & Vogel, 2007 [1986], p. 188). It is also characterized by *Final Lengthening*: its final one or two syllables tend to be lengthened, particularly before a pause. Certain syntactic constructions typically correspond to IPs: notably parenthetical expressions, non-restrictive relative clauses, vocatives, dislocations (Mertens, 2019, p. 92, 97-98; Nespor & Vogel, 2007 [1986], p. 188).

In (38), the bracketed IPs do not belong to the root sentence with which they are associated. Because of them, the remaining prosodic phrases turn into new IPs. For example, (38a) includes three IPs: [Alexander], [the most generous of all kings], and [conquered Darius]. (38f) emphasizes the fact that these IPs are not necessarily syntactic constituents (“Musical Textsetting is” is not a syntactic constituent).

- (38) a. Alexander [the most generous of all kings]_{IP} conquered Darius.
- b. Man [who is mortal]_{IP} is endowed with reason.
- c. It is a great book [isn't it?]_{IP}
- d. [John]_{IP} you should start a new job.
- e. He's so smart [this man]_{IP}
- f. Musical textsetting is [as you know]_{IP} a thorny problem.

Enumerations also produce IPs (Nespor & Vogel, 2007 [1986], p. 201). In “John, Martha, Eric, come here!”, there are four IPs: [John]_{IP} [Martha]_{IP} [Eric]_{IP} [come here]_{IP}.

4.2. Musical Textsetting

The independence of text and music is shown by the existence of stanzaic songs: from one stanza to another, music remains (more or less) stable while the words change (deCastro-Arazola et al., 2015, p. 51). “The brain’s music system appears to operate with functional independence from the language system—the evidence comes from many case studies of patients who, postinjury, lose one or the other faculty but not both” (Levitin, 2006, p. 124-125). A recent fMRI experiment on individuals with or without

severe aphasia confirms that music perception is completely independent from the language system (Chen et al., 2023). This leads to the question of *musical textsetting*. How do we attune the lyrics to the music when we sing or when we listen to a song? How can we have intuitive judgments about the well-formedness of a text-to-tune association? In other words: in song processing, how do we associate the module dedicated to music with the module devoted to language?

Thus far, we have seen correspondence grammars working within the music module. In the case of chant or song, we can posit new correspondence grammars, relating (parts of) the music module with (parts of) the language module. These grammars are necessary to associate linguistic elements such as syllables, moras, linguistic tones, or prosodic phrases with music elements such as meter positions, musical tones, or rhythmic groups. Some properties such as linguistic prominence, linguistic groups, or syllable duration are involved in textsetting.

In chants and in songs, musical tones are not associated with syllable onsets, but with *P-centers* that coincide roughly with the beginning of syllable nuclei (Ryan, 2022, p. 573-574). This is not represented in the grids, which approximately link the musical tone with a full syllable (or with a mora).

4.3. The Text-to-Rhythm Alignment Grammar

In most of the world's song traditions, music groups coincide widely with linguistic constituents (Katz, 2022, p. 16). These linguistic constituents are prosodic constituents (Dell, 2015; Hayes, 1989; Hayes & Kaun, 1996).

If we chant a nursery rhyme on the rhythm of another one, the result is generally awful. (39) (from Baucomont et al. 1970 [1961], p. 31) and (40) are excerpts from two French nursery rhymes.

(39)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
R	•	•	•	•	•	•	•		•	•	•	•	•	•	•	
T	Qu'y	a-	t-il	par-	mi	ces	gens		c'est	l'en-	fant	d'un	grand	gé-	ant	
	<i>What is there among these people</i>								<i>That's the child of a big giant</i>							
	(from Baucomont <i>et al.</i> 1970 [1961], p. 31)															

(40)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
R	•		•		•		•	•	•	•	•		•		•		•		•	•	•	•	•	•
T	Un	deux	trois		nous	i-	rons	au	bois		Quatr'		cinq		six		cue-	illir	des	ce-	ris'			
	One two three we will go to the wood												Four five six picking cherries											

(41) is a gloss of (40).

(41)	[Un]	[deux]	[trois]	[nous	irons]	[au	bois]
	one	two	three	we	go-FUT	to.ART	wood
	One two, three, we will go to the wood						
	[quatre]	[cinq]	[six]	[cueilli-r]	[de-s	cerise-s]	
	four	five	six	gather-INF	ART-PL	cherry-PL	

Four, five, six, to gather cherries

In (39) and (40), the grid is reduced to the rhythm and the linguistic material. The words sit on row T, where T stands for “Text”. The dotted line between row R and row T marks the boundary between the music module (above the dots) and the language module (below the dots).

In *(42), the beginning of the text from (40) is aligned on the musical rhythm from (39). The result grates on the ear, because “au” and “des” are not ending linguistic phrases.

*(42)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
R	•	•	•	•	•	•	•		•	•	•	•	•	•	•	
T	Un	deux	trois	nous	i-	rons	au		bois	quatr'	cinq	six	cue-	illir	des	

Dell and Halle (2009, p. 64-66) have already given a song example of that kind of grouping mismatch. A chant example is important to emphasize that the mismatch only concerns the alignment of the text with the musical rhythm, with no reference to musical tones.

To my knowledge, up to this day, no general principle or particular rule have been proposed to predict the limits of possible grouping mismatches (I leave aside NO RUN-ON, proposed by Hayes, 2009a, and *line/mgroup* proposed by Dell, 2015, p. 199, because they are built on the notion of *line*, a notion I do not use). A constraint such as (43) probably applies.

- (43) GROUPCORR: The edge of a musical rhythmic group matches minimally with the edge of a phonological word.

To test the robustness of (43), one should work out under which conditions it can be violated in a given tradition and, from there, it should be included in a set of constraints that make the *Text-to-rhythm* correspondence grammar. One can imagine, for example, a constraint aiming to relate a group of musical groups with a phonological phrase (ϕ) or with an intonation phrase (IP). Dell (2015, p. 224-225) shows that something of that kind is involved in traditional French songs. Another constraint that probably comes into play concerns the frequency and/or proximity of these discrepancies. A discordance may be accepted when it is isolated within a stanza or when it is separated from other discrepancies by a minimum of concordant rhythmic associations. To be more efficient, (43) could be included in a family of constraints, each constraint dealing with a different layer of prosodic hierarchy.

(43) may be violated also for intentional reasons. In smutty songs, for example, it is common to find music group boundaries that suggest a lewd word that is not officially there. The desired effect is conscious and cannot be explained only with a cognitive grammar. As an anonymous reviewer points out, that fact that this constraint may be violated to produce a comic effect provides evidence of its robustness.

For now, my goal is not to give the exact content of the *Text-to-rhythm* grammar, but to argue that such a grammar has an independent existence and is part of the cognitive system of chant.

4.4. *The Text Associated with Musical Metric-Rhythmic Structure*

Unlike the group correspondences, the associations of linguistic and musical prominences have been extensively studied, mostly in song (rarely in chant). Nevertheless, the role of musical tones and musical tone groups has often been neglected, and the generalizations that have been proposed generally work for chant.

There are different ways to study the matching of prominences in chant or in song. Halle and Lerdahl (1993), Halle (1999), and Hayes (2009a) choose to study a problem of production that is raised by stanzaic songs: when singers know how to align the text with a tune for the first stanza of a song (because this alignment is indicated in a score) how do they find the accurate alignment for the following stanzas? In other words: knowing the words and the musical meter, how do they find the correct rhythm? Perception raises a different problem: how do we mentally find the correct meter when we hear a chant, i.e., a given musical rhythm combined with words? How can we instantly detect ill-formed *Text-to-music* alignments? The two problems must be carefully distinguished, even if with respect to the well-formedness judgments involving textsetting they have much in common.

In song, what is usually called a *line* or a *quatrain* is a string of text which is uttered in a musical chunk. In the present paper, *chunks* are purely musical entities (about chunks in cognitive science; see for example, Gobet et al., 2001; Mathy & Feldman, 2012; Rosenberg & Feigenson, 2013). Chunks are based on musical hypermeters and are built by the mind in working memory with the help of rhythm, melodic contour, and, eventually, the text (about hypermeters, see Halle, 1999, p. 12). In song, the special linguistic properties of the text, usually seen as produced by the use of a prosodic meter, may be seen as a joint consequence of musical chunks and musical textsetting (I will not develop this idea further in the present paper).

Hayes and Kaun (1996, p. 255) made a statistical analysis concerning the alignment of syllabic prominences with metrical prominences in the Appalachian English folklore. They distinguish three levels of strength for metrical positions and four degrees of prominence for linguistic stress. An absence of syllable is treated as a zero degree of stress. In Hayes and Kaun's corpus, a music chunk contains four instances of the *baca* musical meter. Their results are as follows: inside a given musical chunk containing four strong positions, the strongest positions generally match the most accented syllables. The most accented syllables match primarily the fourth strong position, and secondarily the second strong position. The third strong position matches a syllable that is generally more accented than the one associated with the first strong position. The same general frame applies for medium positions; the most accented syllable is the one associated with the fourth medium position, then follow the syllables associated with the second, the third and finally the first position, as shown in figure (44).

(44) *Average linguistic prominence for each metrical position in an American English folk corpus (adapted from Hayes & Kaun 1996, p. 255)*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c			x				x				x				x	
b	x		x		x		x		x		x		x		x	
a	x 	x	x 	x	x 	x	x 	x	x 	x	x 	x	x 	x	x 	x
<i>syllabic prominence</i>	1		5		3		7		2		6		4		8	

A longer line and a bigger bold digit express an average stronger syllabic prominence.

The constraint that is called MATCHSTRESS by Hayes and MacEachern (1998) requires for a stressed syllable to be associated with a rhythmic event itself associated with a strong grid position. Because in chant there is always a one-to-one correspondence between a rhythmic event and a syllable onset (there are no melismata), we can say that MATCHSTRESS requires for a stressed syllable to be associated with a strong grid position. This is a simplification.

Dell and Halle (2009) and Halle (2008) have tried to specify under which conditions stressed syllables are or are not associated with strong positions. But the association of a text with a metric-rhythmic musical structure does not only involve prominence correspondences. Hayes (2009a) shows that other goals are at work. Within a same chunk, big gaps between two syllables/rhythmic events are avoided. Hayes expresses that tendency with the constraint No LAPSE. The local use of extraweak positions (associated with syllables/rhythmic events) is also avoided. Hayes formalizes this goal by a constraint he names *FILL EXTRAWEAK (read as: “Do not fill extraweak [positions]”).

In French, the association of stressed syllables with strong metrical positions is strict at the end of musical chunks (Dell & Halle, 2009, p. 72, say it is strict at the end of a “line”. What they call a line is seen here as a string of text associated with a musical chunk). Anywhere else, this correspondence is just a tendency (Dell & Halle, 2009, p. 73), but this tendency is significant (Temperley & Temperley, 2013, p. 523-524). Furthermore, a lexically stressed syllable is generally located at a stronger position than its adjacent syllables (Temperley & Temperley, 2013, p. 524).

In (22), we have seen how a musical group can be feminine or masculine. In the same way, in the dialect used in traditional French songs (which is not standard French), polysyllabic word-forms and polysyllabic prosodic phrases can be said to be feminine or masculine, depending on their ending. They are feminine when their last vowel is a [ə] (a “schwa”); otherwise they are masculine. A feminine word-form is potentially paroxytonic (= can be stressed on its next to last syllable) and a masculine word-form is potentially oxytonic (can be stressed on its last syllable). I assume that a word-form is necessarily able to be stressed in this dialect, which helps to explain the morphological

difference between feminine word-forms and masculine word-forms.⁹ Analogously, a linguistic phrase can be said to be feminine or masculine, depending on its last vowel. This is summarized in (45).

- (45) *Accentuation of polysyllables in French.* In the French dialect used in folk songs, a polysyllabic word-form is accented on its penult if the nucleus of the last syllable is a schwa; otherwise it is accented on its last syllable.

Some word-forms are always masculine, but other ones can be freely feminine or masculine. In their written form, the latter generally end with the letter *e*, which can be followed by letters *-s* or *-nt*. Whether the word-form is masculine or feminine depends on the performed pronunciation: *cadette* pronounced [ka.'dɛ.tə] is feminine, pronounced [ka.'dɛt] it is masculine; the same holds for *jolie*, which can be feminine ([ʒo.'li.ə]) or masculine ([ʒo.'li]). For a same word-form, a particular pronunciation can be chosen to satisfy textsetting requirements.

In French chants and in French folk songs, the association of music with feminine/masculine linguistic phrases is constrained by (46).

- (46) *Gender Equivalence.* The metrical position associated with the last stressed syllable of an IP is not weaker than the metrical position associated with the pretonic syllable and is not weaker than the metrical position associated with the eventual post-tonic (feminine) syllable.

(47) is made of three IPs, all of them accented on their last syllable. (46) explains why (47) is well-formed, while *(48) is not: the last accented syllable of the last IP in *(48) is *-dai*.

- (47) *A street chant from the protests in Paris in May 1968*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
b	x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x
R	•	•	•		•	•	•		•	•	•	•	•	
T	[E-	tu-	dians]		[Ou-	vri-	ers]		[So-	li-	da-	ri-	té]	
												[ʁi.	'te]	

Students, workers, solidarity!

9 This situation is similar in Southern French, except that the last syllable of a feminine word-form in Southern French cannot be onsetless (in song, *vie* can be disyllabic ([vi.ə]), it can only be monosyllabic in Southern spoken French ([vi])).

*(48)	1	2	3	4	5	6	7	8	9	10	11	12	13	14
b	x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x
R														
T	[E-	tu-	dians]		[Ou-	vri-	ers]		[Sont	so-	li-	dai-	res]	
											[li.	'dɛ.	ɪə]	

Students, workers, [they] do stick together!

Musically, in (46), it is the strength of the metrical positions attached to rhythmic events that is relevant. Musical rhythmic groups are not necessarily relevant. In other words, (46) is part of a correspondence grammar involving the text and a metric-rhythmic structure (it cannot be the *Text-to-rhythm* grammar).

Prominence correspondences involving stressed and unstressed syllables have been studied in a few other traditions: Spanish folk songs (Morgan & Janda, 1989, p. Rodríguez-Vázquez, 2006, 2006–2007, 2010a, 2010b), Italian folk songs (Proto & Dell, 2013), Dutch folk songs (deCastro-Arrazola, 2018, p. 119-138; deCastro-Arrazola et al., 2015), German folk songs (Proto, 2018), and Kaytetye folk songs (San & Turpin, 2021).

Concerning syllable duration, Hayes and Kaun (1996), proposed the “rule schema” (49).

- (49) *Syllable Duration*. Reflect the natural phonetic durations of syllables in the number of metrical beats they receive. (Hayes & Kaun, 1996, p. 260)

In English, a vowel is more or less long depending on its melody (for example, an [ī] is shorter than an [æ], [ōī] is longer than [ī]). Ryan (2022, p. 567-568, 571-578) shows that variations in the phonetic signal (branching onsets and branching codas for example) are statistically reflected in textsetting. As it is written, principle (49) integrates different linguistic phenomena (Ryan, 2022). In particular, we can oppose categorical weight, applying in the syllable domain, and a non-discrete phenomenon such as final lengthening, of which the domain is a major prosodic constituent. In the first case, a metrical position associated with a heavy syllable tends to be followed by an empty metrical position; in the second case, the long final syllable coincides with the right edge of a musical rhythmic group. In both cases, the lack of syllable on a metrical position goes necessarily with the lack of a rhythmic event on that same position (this is true at least for chant and monodic song). What matters is that the long syllable is followed by an *empty* metrical position. Musically, it is not just the metrical position which is at stake. What counts is whether it is filled or not by a rhythmic event. In other words, the textsetting constraint belongs to a grammar involving the whole metric-rhythmic structure in music, and not just the musical meter, as (49) might suggest.

The musical treatment of final lengthening is visible for example in (50), a street shout from the times of the Vietnam War.

(50)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
b	x		x		x		x		x		x		x		x		x
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
R																	
	•		•		•		•		•	•	•	•	•	•	•		•
T	One		two		three		four		We	don't	want	your	fuck-	ing	war		One...

In this example, the word “war” is ending an IP, and it is associated with a rhythmic event which is followed by an empty beat. Note that position 16 is empty linguistically *as well as musically*. There are two different phenomena observed at the same time in this example. Position 16 may be empty for two different reasons: 1) because of (49); and 2) because a linguistic group is associated with a musical group from positions 9 to 15, and this string of sound events is a group because positions 8 and 16 are empty.

In Italian, a stressed syllable is phonetically long. Because of (49), a stressed syllable in a proparoxytone word has to be followed by an empty metrical position (Proto, 2015, p. 124; Proto & Dell, 2013, p. 113-114). This explains why (51a) is well-formed, while *(51b) is ill-formed.

(51a)	1	2	3	4	5	6	7	8	9	10	11	12
c				x						x		
b	x			x			x			x		
a	x	x	x	x	x	x	x	x	x	x	x	x
R												
	•	•	•	•		•	•	•	•	•		•
T	lég-	gi	le	fá-		vo-	le	nel	let-	tí-		no

Read the fairytales in the little bed

*(51b)	1	2	3	4	5	6	7	8	9	10	11	12
c				x						x		
b	x			x			x			x		
a	x	x	x	x	x	x	x	x	x	x	x	x
R												
	•	•	•	•	•		•	•	•	•		•
T	lég-	gi	le	fá-	vo-		le	nel	let-	tí-		no

In *(51b), the stressed syllable “fa-” is on position 4. Position 5 is associated with the syllable “-vo-.” This leaves no empty space after the stressed syllable. This is why the textsetting is ill-formed in this example. (51a) does not have the same defect and is well-formed.

In English, the weight of syllables impacts their association with music (see Hayes & Kaun, 1996, p. 263-267; Ryan, 2022, p. 560-561, 566-567, 574-576).

Concerning mora-timed languages, a general tendency to associate one mora to one metrical position has been observed in various folk traditions: Chuuk and Pohnpei in Caroline Islands (Fischer, 1959) Luganda (Katamba & Cooke, 1987, p. 65), Hausa (Hayes & Schuh, 2019; Schuh, 1988, p. 219-220, 225, 231-232), Somali (Banti & Giannattasio, 1996, p. 92), Japanese (Hayes, 2008, p. 7), Ngizim (Schuh, 2011, p. 227),

and Warlpiri (Turpin & Laughren, 2013, p. 413, 415). Dell and Elmedlaoui (2008) do not make such an observation in Tashlhiyt Berber but the corpus extracts given in their book (2008, p. 183-254) show a clear tendency to equate a mora with a metrical position.

(52) is a *masafo* Somali example (adapted from Banti & Giannattasio, 1996, p. 93-94; the melodic structure is ignored). The *masafo* is a genre of monodic song in Somali folklore. To fully understand the parameters of alignment in this example, one should know that in Somali only the length of the vowels intervenes in syllabic weight (the possible consonantal codas do not add a mora to the syllabic structure). In the example, the long vowels are surmounted by a macron, the short vowels are left without a diacritic. Thus, in this example, the syllables [ba] or [wad] are light, but [lē] and [bān] are heavy (for details, see Banti & Giannattasio 1996, p. 86-87).

(52)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
b			x			x			x			x			x			x
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
R	•	•	•	•	•	•		•	•	•	•	•		•	•		•	•
T	Da-	ba-	ta-	da	ad	shēg-		ta	na	dac-	wad	bān		ka	lē-		ya-	hay
As to the raiders of whom you talk, I also have a complaint																		

In this example (52), I have omitted gridline *c* because my source does not allow me to predict the projection from gridline *b* to gridline *c*; this is not needed for my analysis as we observe that heavy syllables, always projected at least to gridline *b*, are followed by an empty metric position, which suggests a regulated correspondence between the mora and the basic metrical position (that of level *a* in the grid).

Observations like these suggest that (49) must be extended to integrate mora duration. This is proposed in (53).

- (53) *Duration of elementary prosodic units.* Reflect the natural phonetic durations of elementary prosodic units (moras or syllables) in the number of metrical beats they receive.

5. Musical Textsetting in Monophonic Song

When monophonic music has at the same time a text and a melody, it is necessary to try to understand how musical tiers such as meter, rhythm, and tones interact with linguistic tiers such as phrases and syllable nuclei. The present section introduces this problem.

5.1. Tonal Groups, Cadence and Melodic Accent aligned with the Text

We have seen in §.3.1 and §.3.2 that the melodic contour may be important to form musical groups. But the constraints given in (27) are not sufficient to explain all of the possible tone groups. In song, linguistic structure (reinforced by the music's meter and rhythm) may help to discriminate tonal groups. This can be exemplified by (54).

(54a) Score

Sa-vez - vous plan-ter les choux à la mo-de à la mo-de Sa-vez -
 6
 vous plan-ter les choux à la mo-de de chez nous

(54b) Grid

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c			x				x				x				x	
b	x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	E	E	C#	C#	B	C#	A	-	G#	A	B	B	A	B	C#	A
R	•	•	•	•	•	•	•		•	•	•	•	•	•	•	•
T	Sa-	vez	vous	plan-	ter	les	choux		à	la	mo-	de	à	la	mo-	de
	Do you know how to plant cabbages								according to according to							

	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
c			x				x				x				x	
b	x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	E	E	C#	C#	B	C#	A	-	G#	A	B	D'	C#	B	A	-
R	•	•	•	•	•	•	•		•	•	•	•	•	•	•	•
T	Sa-	vez	vous	plan-	ter	les	choux		à	la	mo-	de	de	chez	nous	
	Do you know how to plant cabbages								according to our local custom							

(54) is the refrain of a French nursery rhyme (Robine, 2021, p. 755). On positions 1–7 and 17–23, tonal groups, rhythmic groups, and linguistic groups are perfectly parallel (they are associated with metrical positions having the same strength). Between positions 9 and 16 the phrase “à la mode” (lit., “[according] to the custom”) is repeated. This repetition produces *de facto* the identification of two linguistic groups. These groups are aligned with the musical grid in the same way; this parallel alignment strengthens the linguistic parallelism. On positions 9–12 and positions 13–16, the tone contour is initially rising ($G\#-A-B$ and $A-B-C\#$). In the first case, the group ends with a suspensive tone (B); in the second case, it ends with the tonic (A). The melodic contour is consequently compatible with the parallel linguistic groups. In other words, the identification of two different tonal groups on positions 9–12 and 13–16 is facilitated by the interface of the tone tier with the linguistic tier.

On a larger scale, the tonal group positions 25–32 duplicates this relational schema: it begins like the tonal group located at positions 9–16, then introduces a final variation, ending with the tonic. The whole refrain parallels positions 1–16 with positions 17–32: linguistically, rhythmically, and melodically, two groups are opposed, identical at their beginning, with a variation at the end.

Cadence and melodic accent seem to play a role in musical textsetting, but the problem would deserve further research. Sārg (2006) and Sārg and Ambrazevičius (2007) reveal that in Estonian old folk song *regilaul* singers tend to synchronize melodic accents with lexically stressed syllables. In her study of German *Volkslieder*, Proto (2018, p. 553–554) introduced tonal variations in a well-formed textsetting. Her results suggest that special melodic conditions, including cadence and melodic accent, contribute to the acceptability of a configuration in which a paroxytone word is right-aligned with two metrical positions, the last of which is stronger.

5.2. *Melisma*

A phenomenon specific to song is the melisma. Formally, a melisma can be defined as in (55).

- (55) *Melisma (definition)*. A *melisma* is a set of two or more consecutive rhythmic events which satisfies the following conditions: 1) in this set, the only rhythmic event to be associated with a syllable onset is the first one; 2) every two consecutive rhythmic events belonging to this set must be associated with tone events different in pitch.

In (56b), the beginning of a French nursery rhyme, the melisma is represented by the fact that the rhythmic event on position 11 is not linked with the text. Although melismata are common, they do not occur in all singing traditions. They are not used for example in the Warlpiri songs studied by Turpin and Laughren (2013).

In French, a melisma associated with the next-to-last syllable of a feminine linguistic group (as in [56]) makes it possible to sing a melody that is normally specific to masculine linguistic groups (Dell, 2015, p. 205–207). Compare (56) to (57).

- (56a) *The beginning of a French nursery rhyme. Score*



(56b) *The beginning of the same nursery rhyme. Grid*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c	x				x				x				x			
b	x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	D'	-	-	D'	D'	-	D'	-	D'	-	B	-	G	-	-	-
R	•			•	•		•		•		•		•			
T	Bon-			jour	ma		cou-		si-				ne			
							[ku.		'zi.				nə]			

Hello my cousin-F

(57) *Construct*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
c	x				x				x				x			
b	x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	D'	-	-	D'	D'	-	D'	-	D'	-	B	-	G	-	-	-
R	•			•	•		•		•		•		•			
T	Bon-			jour	mon		très		cher		cou-		sin			
											[ku.		'zẽ]			

Hello my very dear cousin-M

The syllable onset of the melisma in the third measure in the French sample (56) is associated with a rhythmic event itself associated with position 9 in grid (56b). That kind of melisma has interesting consequences on text-to-tune alignment in French folk songs. Indeed, it is possible to have a masculine ending in the text of this sequence, as shown in (57). But note that in (57) the tone linked to position 11 is associated with a syllable onset, while the same note was not associated with any linguistic event in (56). From (56) to (57), the musical part of the song is not changed, the changes take place in the text and in its alignment with musical rhythm (*Text-to-rhythm* alignment). By grid conventions, (56b) is equivalent to the spreading of syllable “si” to the rhythmic event in position 11.

Less often, a melisma sitting on the end of a musical chunk may involve a text with a masculine ending, as in (58).

- (58) *La percheronne*, extract, Voinet (s.d.),
<https://www.partitions-domaine-public.fr/pdf/14520/Traditionnel-La-porcheronne.html>

	1	2	3	4	5	6	7	8	9	10	11	12
c		x				x				x		
b		x		x		x		x		x		x
a	x	x	x	x	x	x	x	x	x	x	x	x
P	A	A		G#	F#	E		F#	G#	A		
R	•	•		•	•	•		•	•	•		
T	Ne	sait		pas		s'ha-		bi-		ller		
								[bi.		'je]		

Does not know how to dress

In (58), the onset of the penult in a *masculine* phrase is associated with the antepenultimate rhythmic event of the chunk. On the basis of observations exemplified by (56) and (58), generalization (59) can be proposed.

- (59) *Melismata in traditional French songs*. Let a masculine rhythmic group be the last in a musical chunk. In traditional French songs, when a melisma ends with the penultimate event of this group, the first rhythmic event involved in the melisma is associated with the penult of a (masculine or feminine) linguistic phrase.

Generalization (59) works whatever the length of the melisma. If melismata of type (56) are more common than type (58) it is probably because type (56) is the only technique authorizing the alignment of a feminine linguistic ending with a masculine musical ending. Masculine linguistic phrases do not need a melisma to be associated with a masculine musical group.

Concerning melisma, Dutch folk songs are more permissive than traditional French songs. deCastro-Arrazola (2018, p. 127-128) revealed from a database analysis that, in case of melisma, a post-tonic syllable can be on a *stronger* position than the stressed syllable.

A melisma may have a very different function in a language like Tashlhiyt Berber, a language that allows syllables bereft of any voicing, whose nuclei may be voiceless plosives. How are such syllables sung? (60) shows two parallel musical chunks that occur in a same song. The lyrics differ but they are sung to the same tune.

(60) *Melisma and rightward shift in Tashlhiyt Berber (from Dell 2011, p. 187 and Dell & Elmedlaoui 2008, p. 204)*

		1	2	3	4	5	6	7	8	9	10	11	12	13	14
c				x				x				x			
b		x		x		x		x		x		x		x	
a		x	x	x	x	x	x	x	x	x	x	x	x	x	x
P		F#	B	B	E	F#	G#	B	B		C#'	D#'	C#'	B	C#'
R		•	•	•	•	•	•	•	•		•	•	•	•	•
T	a.	yan	mu	ta-	ka	ta	ri	sa	wal		sl	žih		tn	nun
	b.	ix	ku-	ri	ni	na	xk	ni-	niz		ra	yad		ra	snt
	a.	<i>Those to whom you give (money), they take your side when they speak</i>													
	b.	<i>If they don't love you, anywhere they see you they lower their eyes</i>													

In (60b), how to sing the syllable associated with position 6? One solution is to perform the syllable without singing it. In perception, the unrealized note “is presumably reconstructed by the listener’s mind” (Dell, 2011, p. 187). Another solution is to unlink from the grid the problematic syllable and to perform a melisma on the preceding syllable. Dell calls this technique a *rightward shift*. This process is represented in (61). On the left is the underlying structure, on the right the surface structure. In the surface structure, the music is unchanged and syllable [xk] is still there. The difference with the underlying structure is that syllable [xk] is not linked to the grid. As a consequence, the musical tone associated with position 6 is now the second part of a melisma, and that melisma is associated with syllable [na].

(61)		5	6	7			5	6	7
	c			x					x
	b	x		x			x		x
	a	x	x	x			x	x	x
	P	F#	G#	B	⇒		F#	G#	B
	R	•	•	•			•	•	•
	T	na	xk	ni-			na	xk	ni-

In Japanese, a mora-timed language, the rightward shift’s melisma is just one of the techniques used to give to a mora the metrical position it needs (see §.4.4). Compare (62) and (63). The word *i-t-ta* (positions 23–25) is not attuned in the same way in these two versions. In (63), there is a melisma and a rightward shift, as in Tashlhiyt Berber; in (62), position 24 is just empty (silent). But in both versions, metrical position 24 is available to hold the mora *t-*. In the (62) rendition, it is crucial that the singer stops to sing mora *i-* before the coming of position 24. Position 24 remains free, and the hearer can mentally associate the unsingable mora *t-* to this position. In the (63) version, the mora *t-* is associated with position 24 *via* a music note (a rhythmic event coupled with a musical tone).

(62) *A sample from the lullaby Edo no komori uta (version sung by Sakura 2004)*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
c	x				x				x				x				x			x				x						x		
b	x	x		x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	A	-	-	-	A	-	G	-	A	C'	-	A	-	G	-	E	G	-	C'	-	A	G	-	-	-	-	-	-	-	-	-	-
R	•				•	•			•	•	•	•				•	•		•		•		•									
T	Bo-	o-		ya		no				o-mo-		ri		wa		do-ko		e		i-	t-	ta										

Where has your nurse been, little boy?

(63) *The same extract from the lullaby Edo no komori uta, Groléziat's (2007) version*

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
c	x				x				x				x			x			x			x			x					x		
b	x		x		x		x		x		x		x		x		x		x		x		x		x		x		x		x	
a	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
P	Ab	-	-	-	Ab	-	G	-	Ab	C'	-	Ab	-	G	-	Eb	G	Ab	C'	-	Ab	C'	G	-	-	-	-	-	-	-	-	-
R	●				●	●			●	●	●	●				●	●	●	●		●		●		●							
T	Bo-	o-		ya		no				o-mo-		ri		wa		do-ko		e		i-	t-	ta										

The technique observed in (63) does not exhaust all of the possible uses of melismata in Japanese. In (63), there is a melisma in positions 19–20, *ko* counts as a single mora and there is no mora associated with position 20.

5.3. Singing a Tone Language

To avoid any ambiguity, in this section I will name *l-tones* linguistic tones and *m-tones* musical tones. The problem concerning the association of *l-tones* with music has been the topic of an abundant literature (see references in Ladd & Kirby, 2020). This literature constitutes a field of its own that seldom interacts with studies dedicated to musical textsetting in general. Exceptions are Proto (2016), McPherson and Ryan (2018), Franich and Lenda Ngemzué (2021), and Kirby (2023). Wee (2007) ignores the musical textsetting tradition but develops a model integrating not only *m-tones* but also the melodic contour and the musical meter.

The treatment of *l-tones* in song is classically tackled via bigrams. A *bigram* consists of a pair of consecutive notes or syllables. In a song, the same note or syllable is examined twice, p. first as the ending of a bigram, secondly as the beginning of the next bigram.

Let us first consider the association of *m-tones* with high or low *l-tones*. There are four possibilities, represented in (64), which is inspired by Ladd and Kirby (2020, p. 678). An underlining indicates a low *l-tone*, and a macron a high *l-tone*.

(64) Bigrams settings between notes and syllables with high or low *l*-tones

a.	b.	c.	d.
ta pā	ta pā	ta pā	ta pā
<i>contrary setting</i>	<i>similar setting</i>	<i>oblique setting 1</i>	<i>oblique setting 2</i>

In (64a) the setting is *contrary* because the second *m*-tone is lower than the first while the second *l*-tone is higher than the first. In (64b), the setting is *similar* because both the melody and the *l*-tones require the voice to go up. In (64c), the setting is *oblique* because the melody is flat while the two *l*-tones are different. In (64d), the setting is *oblique* because the melody goes down while the two *l*-tones are identical.

When the *l*-tones are *rising* or *falling*, there are different theoretical possibilities. For oriental Asiatic languages, the exact quality of each tone is represented by Chao numbers placed between environment bars. Chao numbers label tones on a scale of five levels, from low (1) to high (5). Each tone is described at least by two digits, the first indicating the height at the beginning of the *l*-tone, the last indicating the height at its end. /11/ is a low *l*-tone, /33/ a level *l*-tone, /25/ a rising *l*-tone, and so on. Some *l*-tones having a complex contour are described by three digits. In Mandarin, for example, the third *l*-tone is labelled /214/.

In Cantonese pop songs, data suggest that this is the end point (the target) of each *l*-tone that is relevant (Chan, 1987, p. 30-31). Consequently *l*-tones /35/ and /55/ are similar in song (they are both high); a transition /35–13/ is falling (going from 5 to 3), while /22–35/ is rising and /35–35/ level (high). The association of transition /22–35/ with a rising sequence of *m*-tones is therefore *similar*. The same transition associated with a falling sequence of *m*-tones is *contrary*, while it is *oblique* when associated with a flat sequence of *m*-tones. It is different in Thai pop songs, where *l*-tones /42/ and /24/ both behave as high: for the /42/ *l*-tone, it is the *l*-tone onset that is relevant, while it is the offset for *l*-tone /24/. However, when two /42/ *l*-tones occur consecutively, it is the offset of the second *l*-tone that is relevant (Ketkaew & Pittayaporn, 2014, p. 163-164).

Finally, nine types of settings are possible; they are represented in (65) (Ladd & Kirby 2020, p. 680). The expected frequencies for each type of setting are reported in (66) (Ladd & Kirby 2020, p. 680).

(65) Different possible settings between *m*-tones and *l*-tones

		Melodic sequence		
		Up	Down	Level
<i>l</i> -tonal sequence	Up	Similar	Contrary	Oblique
	Down	Contrary	Similar	Oblique
	Level	Oblique	Oblique	Similar

(66) *Expected frequencies for each type of setting*

<i>l</i> -tonal sequence	Melodic sequence			
		<i>Up</i>	<i>Down</i>	<i>Level</i>
	<i>Up</i>	Frequent	Rare	Possible
	<i>Down</i>	Rare	Frequent	Possible
	<i>Level</i>	Possible	Possible	Frequent

In most of the sung traditions involving *l*-tones that have been studied, the contrary settings are actually rare, as predicted in (66) (Ladd & Kirby, 2020). For example, McPherson and Ryan (2018, p. 127) find in their corpus (songs in Tommo So, a Dogon language), 46.5% of parallel transitions, 48.9% of oblique transitions, and only 3.7% of contrary transitions. In Cantonese pop song, the setting between the *m*-tone and the *l*-tone is similar in 82.66% of cases, oblique in 14.03% of cases, and contrary in only 3.3% of cases (these percentages are inferred from Figure 49.4 in Ladd & Kirby, 2020, p. 682). In another corpus, similar cases go as high as 91.81% (Wong & Diehl, 2002, p. 205). This is the most stringent sung tradition involving *l*-tones that has been described up to this day, aside from Zulu (Schellenberg, 2012, p. 270).

Studies just discussed suggest that there exists a *l*-tones-to-*m*-tones correspondence grammar. But there is also, necessarily, a grammar constraining the association of *l*-tones with the whole musical structure (*l*-tones-to-music). It seems reasonable to think that the two notes of a bigram associated with *l*-tones are not constrained in the same way if they belong to the same musical group or if they do not. This problem is largely understudied. Results may also differ in a significant way depending on whether the syllables involved belong to the same phonological word or not (McPherson & Ryan, 2018, p. 133, 143). It is also possible to think that a given note is more or less constrained depending on its metrical strength. Wee (2007) showed that metrical strength is essential in Mandarin folk songs (see [67d]). The four *l*-tones of Mandarin Chinese are the following ones: 1) the high flat tone /55/; 2) the high rising tone /35/; 3) low rising tone /214/; 4) the low falling tone /51/. (67) suggests that the four Mandarin *l*-tones may be reduced to two tone categories: [+High] and [−High]. The first category covers tones 1 and 2, the second includes tones 3 and 4. Wee (2007, p. 131-134) shows that this bipartite analysis is confirmed by phonetics and phonology. Similarly, Cantonese has six *l*-tones but they are reduced to three categories in song (Schellenberg & Gick, 2020, p. 87).

(67) *Bigrammatic correspondence layout in Mandarin folk songs* (adapted from Wee 2007, p. 131). Let the following notes and syllables be in correspondence:

note 1	note 2
syllable A	syllable B

- If note 1 is higher than note 2, then syllable A bears the first or the second *l*-tone. Syllable B may bear any *l*-tone.
- If note 1 is lower than note 2, then syllable A bears the third or the fourth *l*-tone. Syllable B may bear any *l*-tone.

- c) If note 1 and note 2 have the same pitch, syllable A can bear any of the four *l*-tones.
- d) (a) and (b) are almost always true when note 1 is on a strong position and is in the same measure than note 2.

Medumba is a Grassfields Bantu language spoken in Cameroon. In its folk songs, high *l*-tones are preferably associated with strong metrical positions, low *l*-tones to weak positions, while contour *l*-tones have no preference in this matter; atonal syllables appear exclusively on weak positions (Franich & Lendja Ngnemzué, 2021, p. 21).

Sung *l*-tone texts present interesting phenomena involving melismata. In Thai, melismata are frequently used to match with rising or falling *l*-tones (Saurman, 1999). In Mandarin, when a syllable *n* initiating a melisma is associated with a strong metrical position, the first and the last *m*-tone of the melisma generally differ in pitch in the same way than the two *l*-tones associated with them differ in contour. Otherwise, the [$\pm$ High] contrast between syllables *n* and *n+1* must be reflected by a similar contrast between the last *m*-tone associated to syllable *n* and the first *m*-tone associated to *n+1*. These constraints may be violated in case of positional parallelism (see Wee, 2007, p. 135 for details).

Another problem is the textsetting treatment of pitch accent in a language like Japanese. To my knowledge, Cho (2017) is the only study dedicated to this topic. Cho (2017, p. 33) shows that in Japanese traditional nursery rhymes there is, as in tone languages, a tendency to associate linguistic bigrams with musical bigrams having the same shape (rising, falling, or flat), even if this tendency is weaker than it is in tone languages like Cantonese or Vietnamese. A chi-square test reveals that statistic results are significative ($\chi^2 = 115.57$, $df = 2$, $p < .0001$; Cho, 2017, pp. 33–34). In Japanese, a mora with a high tone may be contrastive (it bears a pitch accent) or may be demarcative (it bears a *boundary tone*, indicating the beginning of a word or a phrase; for details, see Labrune, 2012a, p. 178ff.). When two moras of a linguistic bigram belong to a same word and one of them is accented, the results are stronger: the correspondence is 68.27% for falling bigrams, 56.6% for rising bigrams (Cho, 2017:34–35). There is also a significant tendency to associate moras bearing a pitch accent with strong positions, even if the tendency is weaker than it is in French or in English as regards the association of stress and strong positions (Cho makes her comparison with Temperley & Temperley's, 2013, results).

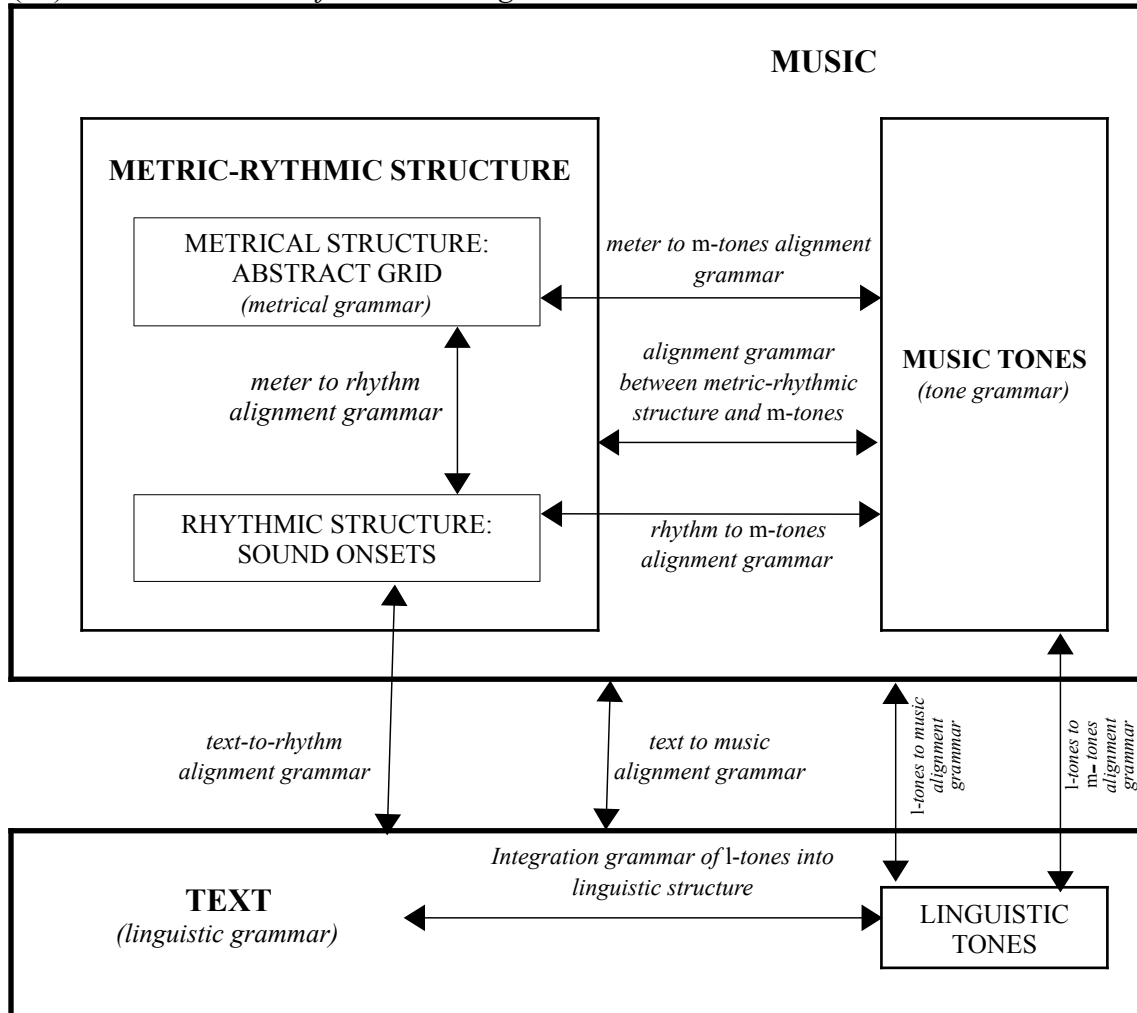
All of the facts discussed here provide strong evidence in favor of the existence of a correspondence grammar involving the musical structure taken as a whole (*l-tones-to-music*, see above), in addition to a *l-tones-to-m-tones* grammar.

6. A Modular Model of Monodic Song

The whole pattern of modules and grammars involved in monodic song that are discussed in the previous sections is represented in (68). Compared to the modular model of music processing proposed by Peretz & Coltheart (2003), (68) does not represent the emotions or the interface with the sensory-motor system but develops the

text to music interface. These subsystems of music are generally ignored in musical textsetting studies.

(68) *A modular model of monodic song*



Previous works focused mainly on song (the most complex form studied in the current paper), with little attention given to chant, amelodic music, or instrumental monodic music. The result is an incapacity to distinguish the different modules and module interfaces represented in (68). Different modules or submodules and different interfaces require different grammars (modular grammars or correspondence grammars). From (68), it should be possible to build a typology of musical forms, with constraint rankings potentially different from one tradition to another.

7. Recapitulation

In this section, I sum up the different modules, submodules, and correspondence grammars that have been distinguished in this work.

- The Metrical grammar, represented at the left top of (68), treats the processing of musical meters. In this paper, musical meters are represented by abstract grids. These grids are built in compliance with a set of constraints (see §.1).
- The *rhythmic structure* submodule operates with a grammar that defines the conditions for some object to be a rhythmic event relevant for music. This grammar also determines the possible structures of rhythmic groups (§.2.1).
- The *Meter-to-rhythm* correspondence grammar regulates the synchronization of a given rhythmic sequence with a meter. It also distinguishes between feminine and masculine rhythmic groups (§.2.2).
- The submodule *music tones* consists of a grammar P which allows the establishment of tonality (modes, scales) and m -tone groups (§.3.1).
- The function of *Rhythm-to-m-tones* correspondence grammar is to associate m -tones and rhythmic events (§.3.2).
- The *Meter-to-m-tones* correspondence grammar determines the association of meter to m -tones. It helps the recognition of modes (§.3.3).
- The alignment grammar between the metric-rhythmic structure and m -tones determines of the *cadence* of a piece of music (§.3.4).
- All of the submodules and correspondence described so far are part of the *music* section of the proposed system.
- The *text* section (§.4.1) is a filter which extracts some linguistic information from the module dedicated to language processing (the language module *per se* is not represented in [68]). This information is related to the *music* section by various correspondence grammars.
- The submodule *linguistic tones* is specialized in the extraction (from language processing) of information concerning l -tones. Structural features of pitch accent may be included in this grammar. This grammar is not developed in this paper.
- Integration grammar *l-tones-to-text*, part of the *text* section, extracts from language processing constraints governing the association of l -tones (or pitch accents) with prosodic constituents such as moras, syllables or phrases. Again, this grammar has to be assumed but is not developed in this paper.
- The *Text-to-rhythm* correspondence grammar links linguistics major prosodic constituents (ω , ϕ , IP) to musical rhythmic groups (§.4.3).
- The *Text-to-music* alignment grammar processes the association of linguistic units such as moras or syllables to metric-rhythmic musical units (strong, medium, or weak rhythmical events, depending of their association with the meter). It contains also more complex operations, such as melismata, which involve m -tones (§§.4.4, 5.1–5.2).
- The grammar *l-tones to m-tones* specifies how to associate linguistic tones with musical tones. These associations have been studied thus far as bigrammatic correspondences (§.5.3).
- At last, the *l-tones-to-music* alignment grammar concerns the correspondence of l -tones with musical structure as a whole (metrical positions, musical groups) (§§.5.3).

8. Hierarchy Between Grammars

It is not apposite to rank constraints that belong to different grammars in a single hierarchy. Nevertheless, some grammars may be privileged. From a certain point of view, the musical module always prevails over the linguistic module in singing. A text, considered in itself, has its own rhythm and its own intonation contour. A melodic musical piece, for its part, has a metric structure, rhythm, and melodic contour. When one puts this text and this music together to produce a song, it is the musical structure that dominates: the text is pronounced according to the rhythm and melody of the music. Nevertheless, the ranking between grammars can vary from one tradition to another, the prosodic structure of the text being more or less well preserved. In Spanish folklore, data suggest that musical structure is clearly prioritized over textual structure (Rodríguez-Vázquez, 2006, p. 278). In traditional French songs, the alignment of text with music is made in a way that preserves the prosodic properties of texts (prosodic groups correspond to musical groups, and masculine linguistic endings are generally associated with music masculine endings). On the contrary, in Tashlhiyt Berber folk songs, music structure is distinctly privileged over linguistic structure (Dell & Elmedlaoui, 2008, p. 67). In English folk songs, prominence correspondences win over group correspondences (Hayes & MacEachern, 1996, §.4). Ranking grammars is a technique that allows these observations to be implemented in a cognitive theory of modular and parallel systems.

Turpin (2005, 2007a, 2007b) studies the *Akwelye*, a sung tradition in Kaytetye, a language from Central Australia. Each *Akwelye* “song item” has its own text (Turpin, 2005, p. 92), “set to an unchanging rhythm”; this text “is repeated until the end of a melodic structure” (Turpin, 2005, p. 98). This repeated text/rhythm, together with its melody, makes a “small song”. Different consecutive small songs make a “song series” (Turpin, 2005, p. 92). A song item is 30 to 40 seconds long and can exceed one minute if it comes with a dance. A small song generally contains between 1 and 5 song items; in a corpus of 227 small songs, 88.1% of them contain between one and five song items, the most frequent proportions being two song items (26.87%), three song items (22.9%) and four song items (19.82%) (calculated from Turpin, 2005, p. 96). A series may contain over eighty song items, and includes short intervals of chatter and preparation between them (Turpin, 2007a, p. 103). Breaks between small songs are much longer (up to five minutes) (Turpin, 2005, p. 95). In *Akwelye*, each song belonging to a same series uses a same repeated melodic structure. In other words, correspondence grammar *Text-to-rhythm* is privileged at the *song item* level, while the *Text-to-music* grammar is favored at the *small song* and at the *series* level. The musical rhythm of a song item is made of rhythmic chunks. There exists 16 possible rhythms for a chunk in the whole tradition (Turpin, 2007b, p. 98-99). A same rhythmic chunk may appear in different song items. Moreover, in *Akwelye* Kaytetye folk songs, each half of a musical chunk is organized in two rhythmic groups, and the first of these groups may not contain less rhythmic events than the second (Turpin, 2007a, p. 100). The text must submit to this music constraint. At that level, the metric-rhythmic musical structure wins over linguistic structure.

These relations of competition between language and music favor the interpretation of music and text as distinct mechanisms (Peretz & Kolinsky, 2009, p. 154).

9. Conclusion and Open Questions

The goal here is to propose a global architecture to musical textsetting. Model (68) aims to make explicit what is at stake in the research dedicated to musical textsetting and to formulate hypotheses that can be tested and eventually falsified. These hypotheses propose various modules, submodules, or correspondence grammars, interacting in a definite way.

The basic idea is that two different objects are associated in song: a tune and a text. Each of these objects is conceived as processed by a specific module of the mind-brain. Neuropsychological data strongly argues in favor of this double dissociation (Deutsch, 2019, p. 140-141, 155-162, 188; Peretz, 2018, p. 89; Peretz & Kolinsky, 2009, p. 147-149). Case studies document accidental or congenital amusias having no consequence on language processing (Fitch, 2006, p. 200; Patel, 2008, p. 4, 72, 230; Peretz & Coltheart, 2003, p. 688-689). Conversely, cases of individuals suffering from aphasia and having no problem in music processing are well documented (Chen et al., 2023; Peretz, 2012, p. 258-259). Recent experiments by functional MRI strongly suggest that musical perception does not require the activation of the linguistic system (Chen et al., 2023).

Each module, submodule, or correspondence grammar in this model should inform the design of future experiments. Empirical data can be interpreted in various ways and cannot, alone, determine the choice between this model and alternative ones. Cognitive experiments seem necessary. For example, Katz and Pesetsky (2011, p. 30ff.) suggest that the formal constraints at play in language and music are the same, the observed differences resulting from the use of lexicon in language and tonal classes and chords in the music. This should lead to a quite different conception of musical textsetting. A non-modular approach to song, to be coherent, would need a theoretical foundation at least as solid as the theory of modularity. In particular, it should show that the processing of rhythm or pitch is produced by the mind-brain with the same tools for both music and for language. Existing observations are not all convergent (see this paper's introduction) but most argue for separate mechanisms. A non-modular approach to musical textsetting would perhaps have the merit of avoiding the theoretical no man's land that are alignment grammars: sometimes they seem to be intra-modular, sometimes inter-modular (as appears in [68]). One might think that correspondence grammars are not modular and relate to so-called "open" or "central" (Fodor, 1983, p. 101-119) cognitive processes. But these grammars are fast operating systems and produce very focused operations, in a very limited "language". In this respect, they function like modules. Adopting a Bayesian approach could perhaps resolve the problem, particularly for *Text-to-rhythm*, *Text-to-music*, *l-tones-to-music* and *l-tones-to-m-tones* grammars; that is, correspondence grammars associating linguistic elements with music elements (see [68]). Indeed, these grammars combine elements that are similar in some way (prominence units, phrases or groups, tones, etc.). Calculating a Bayesian inference could accommodate also the fact that correspondence grammars do not behave in the same way from one cultural tradition to another (compare French folk songs to French

rap for example) and that there seems to be a competition between grammars (see §.8). Further research is necessary to get a better understanding of correspondence grammars and the way they work.

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